APEXPLANET SOFTWARE PVT. LTD. · DATA ANALYTICS INTERNSHIP · TASK 4

End-to-End Sales Analytics,

From Raw Data to Business Intelligence

From 1,000 raw transactions to statistically validated business decisions

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1,000	**₹139.4M**	**4 / 4**
TRANSACTIONS ANALYSED	TOTAL REVENUE	QUESTIONS ANSWERED

PROJECT SCOPE

6
Analytics disciplines

8
Cities covered

5
Product categories

4
Statistical tests

2
Power BI dashboards

8
Action recommendations

One dataset, six disciplines, one decision-ready narrative

An end-to-end engagement: every stage from raw data to validated boardroom story

The objective

Transform a raw sales log into intelligence a business can act on — identifying what drives revenue, who buys, and where growth belongs.

What makes it end-to-end

Six disciplines run in sequence, each feeding the next. Every claim is formally tested before it reaches this deck. Every stage produced a versioned, auditable deliverable.

Six delivered artifacts

Cleaned dataset · SQL library · EDA notebooks · test suite · 2 Power BI dashboards · this narrative

Data Cleaning

Profiling, nulls, types, de-duplication

Exploratory Analysis

Trends, mix, category and city distributions

SQL Intelligence

Business KPIs queried against MySQL

Statistical Tests

t-test, ANOVA, correlation, CI at $\alpha = 0.05$

Power BI Dashboards

Two interactive views for daily use

Executive Story

Insights framed as decisions, not data summaries

Four questions the business could not answer with confidence

The data existed — the structure, evidence and statistical confidence to act on it did not

What drives revenue?

Which categories, products and price points actually move the ₹139.4M top line — and by how much?

Who is the customer?

Do gender and age genuinely shape spending behaviour, or are those assumptions myths waiting to be retired?

Where is the growth?

Which of the 8 cities deserve marketing and inventory investment first, and by what margin?

Can we trust the patterns?

With a 1,000-row sample, are the signals real — or artefacts of noise that evaporate under formal testing?

Decision at stake: next year's inventory levels, marketing allocation and customer retention strategy.

1,000 transactions, audited and analysis-ready

Retail sales transaction log · FY2025 · one row per order line

1,000

TRANSACTIONS

992 unique orders · FY2025

13

ATTRIBUTES PER ROW

order, customer, product

8

CITIES

metro + tier-2 coverage

5

PRODUCT CATEGORIES

Electronics to Grocery

ORDER

Order_ID

Order_Date

992 unique orders

Jan – Dec 2025

grain: one order line

CUSTOMER

Customer_ID

Name

Age · Age_Group

Gender

City

TRANSACTION

Product

Category

Quantity

Unit_Price

Total_Sales

Quality audited before every analysis

Type casting · null handling · de-duplication · outlier review · documented data quality report

Derived field: Age_Group (Young / Adult / Middle Age / Senior) engineered during wrangling.

A production-shaped toolchain — nothing decorative

Each tool earns its place; the full pipeline is reproducible end-to-end on GitHub

Data Engineering

Python — core analysis language

Pandas — wrangling and aggregation

NumPy — numerical computing

Storage & Query

MySQL — relational data store

SQL — business KPI queries

Ingest — cleaned CSV loaded to MySQL

Business Intelligence

Power BI — two interactive dashboards

Data model — KPI cards and drill-downs

Slicers — month, city, category filters

Workflow & Versioning

Jupyter — reproducible notebooks

Git — version control

GitHub — public portfolio repository

Fully reproducible: raw CSV → Python cleaning → MySQL → notebooks → Power BI · all versioned on GitHub

Six stages, each closing with a committed artifact

Insight quality is decided long before the dashboard — discipline is the differentiator

01

Data Immersion

Profile 1,000 raw records; data quality audit; data dictionary with field-level notes.

data quality report

02

Wrangling

Cast types, handle nulls, de-duplicate records; engineer the Age_Group feature.

cleaned_dataset.csv

03

EDA in Python

Revenue distributions, monthly trends, category mix, city and age breakdowns.

EDA notebooks

04

SQL Analysis

Express every business KPI as a MySQL query; reproducible query library committed.

MySQL query library

05

Statistical Validation

t-test, one-way ANOVA, Pearson correlation, 95% CI — $\alpha = 0.05$ throughout.

statistical test suite

06

BI & Narrative

Two Power BI dashboards and this executive narrative close the pipeline.

dashboards + narrative

Every stage closed with a versioned artifact on GitHub — any number on any slide traces back to the raw rows that produced it.

The business on one page

FY2025 headline metrics — the baseline every later insight refines

TOTAL REVENUE

₹139.40M

12 months · 5 categories · 8 cities

TOTAL ORDERS

992

≈ 2.7 orders per day, steady throughout the year

UNIQUE CUSTOMERS

947

95% single-purchase buyers — retention is the largest growth lever

UNITS SOLD

5,435

across 6 product lines, led by Mobile on volume

AVERAGE ORDER VALUE

₹140,524

95% CI: ₹132,319 – ₹146,480

REVENUE PER CUSTOMER

₹147,201

52% male · 48% female — statistically indistinguishable

A resilient year with one addressable soft patch

Monthly revenue, ₹ million · FY2025 · seasonal, not structural decline

MONTHLY REVENUE (₹M)

PEAK MONTHS

Mar ₹13.1M · Jun ₹12.9M · Nov ₹12.6M

Fiscal year-end and festive cycles reliably lift demand.

SOFT PATCH

Aug ₹9.4M · Sep ₹9.2M

A –30% trough vs. the March peak. Seasonal, not structural.

STRATEGIC READ

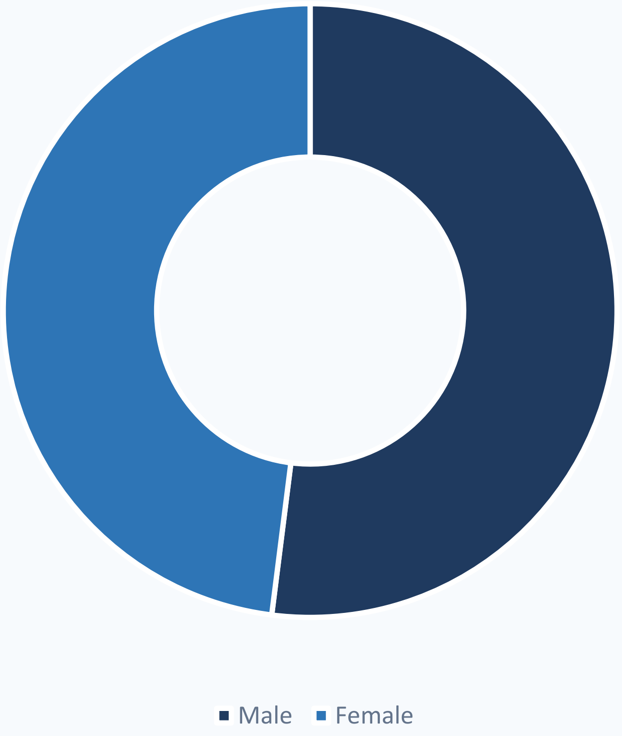
₹9M–₹13M operating band

No downtrend. Fix the trough with counter-seasonal campaigns; protect the peaks with inventory.

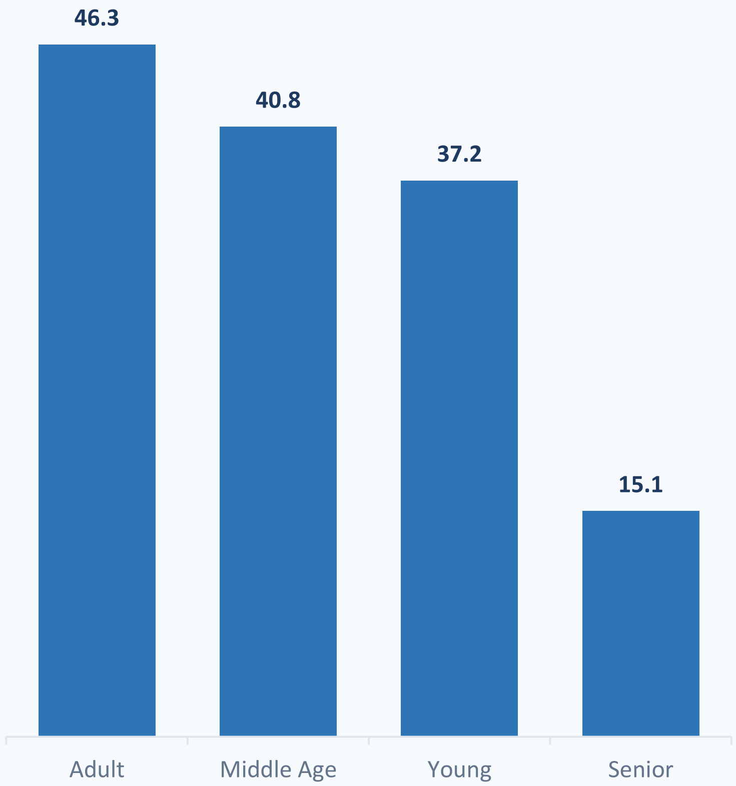
Spending is uniform — demographics are not a segment

Gender and age patterns, then the statistical verdict on both

REVENUE BY GENDER



REVENUE BY AGE GROUP (₹M)



What the tests found

Gender gap is statistical noise

t-test $p = 0.495$ — the 52:48 split does not hold under testing.

Age groups spend identically

ANOVA $p = 0.973$ — seniors buy less often, not smaller baskets.

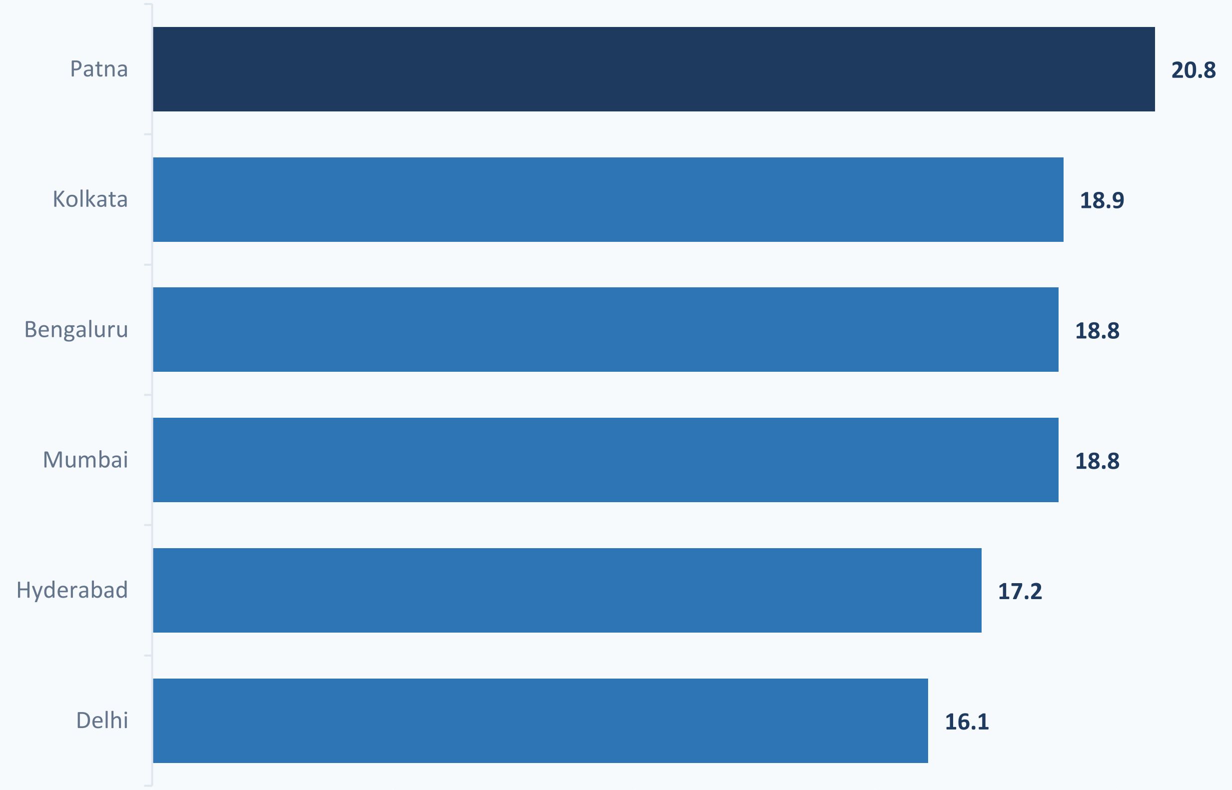
Implication for strategy

Segment on behaviour — category, city, season — not demographics.

Patna leads every metro — tier-2 India buys premium

Revenue by city, ₹ million · 8 markets ranked

REVENUE BY CITY (₹M)



MARKET LEADER

Patna — ₹20.8M

14.9% of national revenue; out-sells Mumbai, Bengaluru and Delhi.

CONCENTRATION

Top 5 cities = 67.7%

Two-thirds of revenue in five markets — efficient focus, manageable tail.

STRATEGIC READ

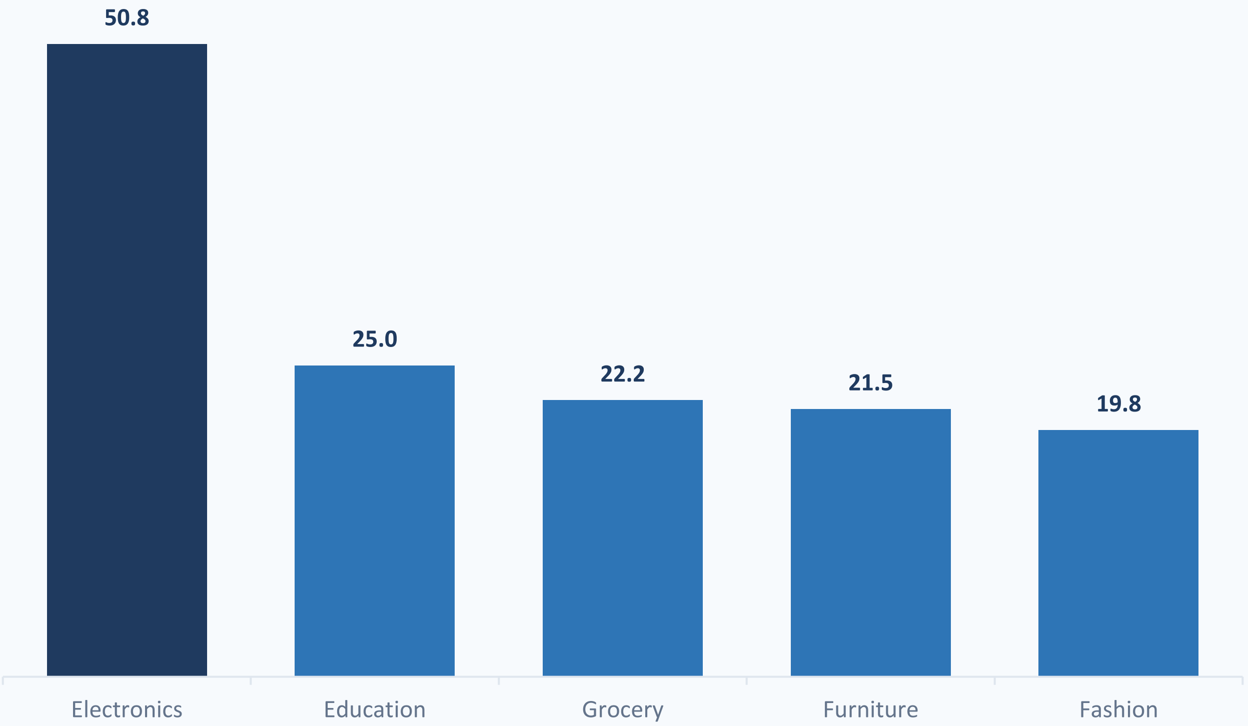
Revenue density, not city size

High-value baskets in tier-2 India. Expand by revenue density, not metro prestige.

Electronics is the engine — Fashion is the repositioning brief

Category revenue, ₹ million · with product-level standouts

REVENUE BY CATEGORY (₹M)



Laptop — ₹25.4M

Highest-revenue product; premium anchor of the Electronics engine.

Mobile — 1,008 units

Highest-volume product; the traffic driver for the category.

Grocery — ₹145.3K AOV

Highest average order value of any category; bulk-buying behaviour.

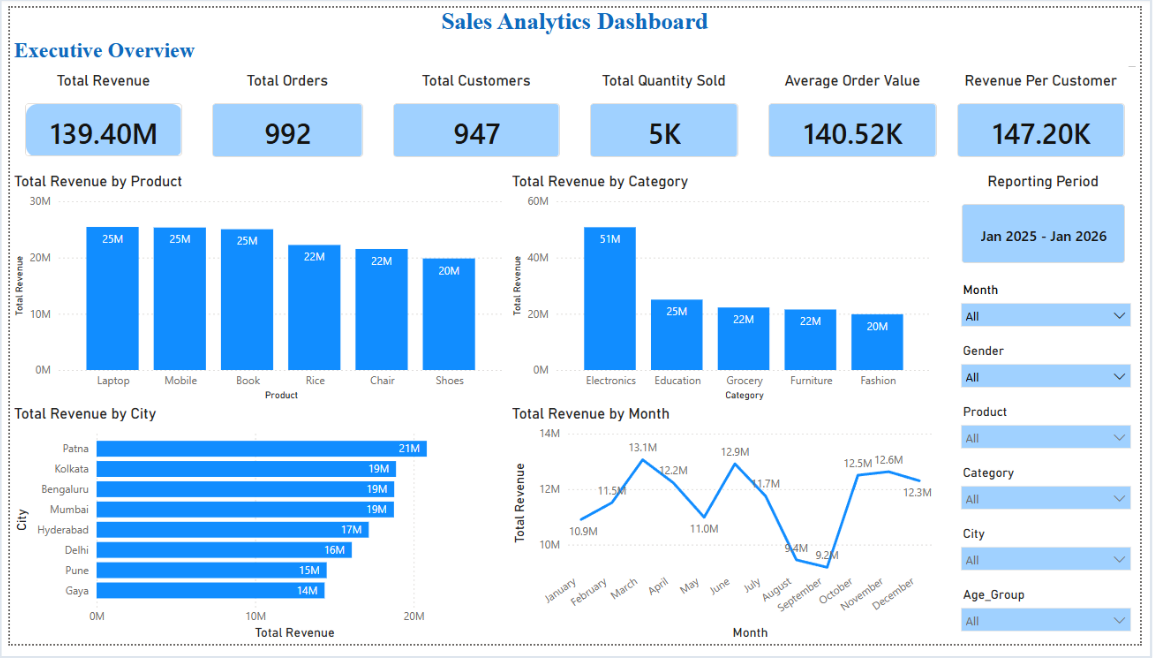
Fashion — ₹19.8M

Weakest line, –61% vs Electronics. Repositioning candidate: pricing and bundle strategy.

Two dashboards: monitor operations, then target customers

Executive Overview for daily posture · Customer Deep Dive for targeting decisions

01 · Executive Overview



KPI cards · revenue by product, category & city · monthly trend

02 · Customer Deep Dive

Gender & age analysis · top customers · orders by category · AOV

Both dashboards are fully interactive — slicers for month, city and category drive every visual simultaneously.

Every reported insight survived a formal test

Four tests at α = 0.05 — only findings that passed are in this deck

PEARSON CORRELATION

r = 0.6466

SIGNIFICANT ✓

Quantity sold and revenue move together strongly. Volume is a proven growth lever — attach-rate programmes and cross-selling are provably profitable, not just directionally sensible.

INDEPENDENT T-TEST

p = 0.4950

NOT SIGNIFICANT

The 52:48 gender split is statistical noise. No meaningful difference in basket size or frequency. Gender-based segmentation wastes budget — redirect to city, category and season.

ONE-WAY ANOVA

p = 0.9727

NOT SIGNIFICANT

Order values are statistically identical across all four age bands. Seniors buy less often, not smaller baskets. Age-targeted messaging is unlikely to shift spend per transaction.

95% CONFIDENCE INTERVAL

₹132,319 – ₹146,480

SIGNIFICANT ✓

The true average order value sits in this band with 95% confidence. Tight enough to build an AOV target against. Any promotion lifting spend above ₹146.5K is measurably above baseline.

Method: SciPy statistical suite on the cleaned dataset · α = 0.05 throughout.

One engine, two gaps, three opportunities — and three threats to manage

Every point traces to a tested finding — this is data, not opinion

S

Strengths

- Electronics engine at 36% of total revenue
- Premium basket: ₹140.5K average order value
- Balanced 8-city footprint across metros and tier-2

W

Weaknesses

- Fashion at ₹19.8M — 61% below Electronics
- Aug–Sep trough: –30% below the March peak
- ~95% single-purchase buyers — near-zero retention

O

Opportunities

- Cross-category bundles to push AOV past ₹140K
- Loyalty programme to convert 947 one-time buyers
- Reinforce Patna & East-India revenue momentum

T

Threats

- Seasonal cash-flow swings from demand troughs
- Revenue concentration: top 5 cities = 67.7%
- Metro price competition on Electronics and Fashion

Eight moves, each tied to a tested finding

Sequenced from quick wins to structural plays — none based on intuition

01 Protect the Electronics engine
Prioritise inventory and availability for the 36% revenue core.

02 Promote Laptop and Mobile as hero SKUs
They lead both revenue (₹25.4M) and volume (1,008 units).

03 Launch cross-category bundles
Lift AOV past ₹140.5K — Grocery's ₹145.3K AOV proves the headroom exists.

04 Build a loyalty programme
Convert the ~5% repeat rate — the single largest untapped lever in the data.

05 Focus on the top-5 cities first
Patna, Kolkata, Bengaluru, Mumbai, Hyderabad = 67.7% of revenue.

06 Reposition Fashion with pricing and bundles
Revival of the weakest line rather than category exit.

07 Systematise cross-selling at checkout
Volume correlates with revenue at r = 0.65 — attach-rate is provably profitable.

08 Counter-seasonal Aug–Sep campaigns
Pre-festive offers and targeted pricing to smooth the –30% trough.

Six outcomes already in stakeholders' hands

Measured against the four original business questions — all answered with quantified confidence

Revenue drivers quantified

36% of revenue traced to Electronics; Laptop + Mobile confirmed as hero SKUs worth ₹50.8M combined.

Costly assumptions retired

2 demographic myths — gender and age — disproven before they misdirected a marketing budget.

Insights statistically certified

4 formal tests at $\alpha = 0.05$ — only findings that survived statistical scrutiny are reported.

Decision tools in daily use

2 interactive Power BI dashboards — executive monitoring and customer targeting, live now.

Evidence-backed action plan

8 recommendations, each traceable to a tested finding — not one based on intuition.

Business questions answered

4 / 4 of the original questions resolved with quantified confidence intervals where applicable.

From raw rows to decisions leadership can sign

Three things this project proves — about the business, and about the method

From raw to reliable

1,000 messy records became a cleaned, documented, SQL-queryable asset with a full audit trail — every number on every slide is traceable to source.

Patterns, verified

Formal testing confirmed the real drivers — volume, category, geography — and retired two myths about gender and age that would have misdirected budgets.

Decisions, operationalised

Two live dashboards and eight evidence-backed recommendations turn analysis into an operating rhythm the business checks daily.

“Good analytics doesn’t end with charts — it ends with decisions someone is willing to sign.”

From describing the past to predicting and automating the future

Six natural extensions — each building on an asset this project already created

Demand Forecasting

ARIMA or Prophet on the monthly revenue series — plan inventory ahead of the March and festive-season peaks now proved in the data.

RFM Segmentation

Cluster the 947 customers by Recency, Frequency and Value to give the loyalty programme a precise, scored target list rather than a blanket offer.

Market-Basket Analysis

Association rules across the 1,000 transactions to design recommended bundles from real co-purchase patterns, not assumption.

Automated ETL Pipeline

Scheduled ingest, clean and load — removing every manual refresh step and enabling daily data freshness in the dashboards.

Real-Time Dashboards

Power BI Service with scheduled refresh and row-level security per region, so each city manager sees only their market.

Promotion Uplift Testing

A/B frameworks to measure campaign ROI as it happens — extending the statistical toolkit from validating the past to optimising the future.

SALES ANALYTICS · END-TO-END PROJECT · APEXPLANET DATA ANALYTICS INTERNSHIP · TASK 4

Thank you.

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Project repository github.com/shettyprasad-git/Sales-Analytics-End-to-End-Project-ApexPlanet

Python · Pandas · NumPy · SQL · MySQL · Power BI · Jupyter · Git · GitHub

PROJECT AT A GLANCE

₹139.4M
revenue analysed

1,000
transactions processed

4
statistical tests run

2
Power BI dashboards

8
action recommendations

4 / 4
questions answered